SUPERSEDING
MIL-R-55344A(EL)
28 May 1975

MILITARY SPECIFICATION

RADIATION SOURCE, INFRARED
FOR
SEARCHLIGHT, INFRARED, AN/VSS-3

This specification is approved for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers one type of radiation source, infrared, (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-71	Pallet, Materials Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial)
QQ-S-781	Strapping, Steel, Flat and Seals
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Shipping, Fiberboard
PPP-F-320	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, (For Carton Sealing)
PPP-T-97	Tape, Pressure-Sensitive Adhesive, Filament Reinforced

MILITARY

MIL-P-116	Preservation-Packaging, Methods of
MIL-M-10304	Meters, Electrical Indicating, Panel Type, Ruggedized, General Specification for

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MIL-P-11268	Parts, Materials, and Processes Used in Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-P-26514	Polyurethane Foam, Rigid or Flexible, for Packaging
MIL-B-43014	Boxes: Water Resistant Paperboard; Folding, Set-Up, and Metal-Stayed

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads 40 Inch x 48 Inch Pallets, Skids, Runners, or Pallet Type Base
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurements of
MIL-STD-810	Environmental Test Methods

DRAWINGS

ELECTRONICS COMMAND

SC-DL-646812	Radiation Source, Infrared

(Copies of specifications, standards, and drawings required by the supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28	Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

3. REQUIREMENTS

3.1 Description. The Radiation Source, Infrared (RSI) shall be a gaseous discharge, short-arc device utilizing xenon gas in a quartz envelope and fitted with electrical connections.

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3.2 Construction. The RSI shall be constructed in accordance with SC-DL-646812 and as specified herein.

3.3 First article. When specified in the contract (see 6.2) the contractor shall furnish first article RSI(s) for inspections (see 4.3).

3.4 Parts and materials. Parts and materials shall be as specified herein and as shown on the applicable drawings. Parts and materials not specified shall be selected by the supplier in accordance with MIL-P-11268 and shall conform to all applicable provisions of this specification.

3.5 Operation and environment.

3.5.1 Quartz bulb. The optical quality of the quartz bulb shall be such that the apparent shift of the cathode observed in any direction shall lie within a circle of 0.5 millimeter radius.

3.5.2 Ignition and operation. The RSI shall ignite and maintain ignition within 3 seconds under all environmental and other conditions specified herein with a driving circuit as specified in Figure 2 and any power supply voltage between 22.0 vdc, plus 0.5 or minus 0.0 volts and 28.0 vdc, ± 0.5 volts during the first 100 hours of RSI operation and 24.0 vdc, ± 0.5 volts and 28.0 vdc, ± 0.5 volts for the remainder of RSI life. The lamp shall have a voltage-current characteristic such that after the lamp has been ignited and thermally stabilized in operation for at least 5 minutes, the voltage at 25 amperes shall be 16.4 vdc, ± 0.5 volts and at 47.5 amperes shall be 19.5 vdc, ± 0.5 volts.

3.5.3 Luminance. After 5 hours of lamp operation, the average luminance of the arc over a circular area 1.0 millimeter in diameter centered on the optical axis of the RSI shall be not less than 840 candelas per square millimeter with the RSI operating with a driving circuit as specified in Figure 2 with a power supply voltage of 28.0 vdc, ± 0.5 volts. After 300 hours of lamp operation under ON-OFF cycled conditions as specified in paragraph 4.6.11, the luminance as stated above shall be no less than 600 candelas per square millimeter.

3.5.4 Mechanical shock. The RSI shall not be damaged (see 6.3) by mechanical shocks of 70 g's, plus or minus 5 g's, with a pulse duration of 6 milliseconds, plus or minus 0.5 millisecond.

3.5.5 Vibration. The RSI shall not be damaged by vibrations along the axis of the lamp electrodes throughout a frequency range of 21 to 26 hertz (Hz) at a value of 16.0 g's, plus or minus 0.5 g, and from 130 to 150 Hz at a value of 8.0 g's, plus or minus 0.5 g. In a direction perpendicular to the axis of the lamp electrodes, the RSI shall not be damaged by vibrations throughout a frequency range of 29 to 34 Hz at a value of 16.0 g's, plus or minus 0.5 g, and from 70 to 90 Hz at a value of 10 g's, plus or minus 0.5 g.

3.5.6 Storage temperature. The RSI shall not be damaged by storage in any ambient temperature from plus 68°C to minus 62°C at all possible relative humidities within that range.

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3.5.7 Operating temperature. The RSI shall perform at any ambient temperature from plus 52°C to minus 54°C with humidity controlled between 85 and 95 percent at temperatures above plus 23°C, ± 2°C temperature (see 6.3).

3.5.8 Altitude. The RSI shall not be damaged by ignition and operation at 10,000 feet above sea level at a temperature of plus 35°C.

3.5.9 Altitude transportation. The RSI shall not be damaged by transportation in unpressurized aircraft at an altitude of 40,000 feet above sea level at a temperature of minus 62°C.

3.5.10 Electromagnetic compatibility. The electromagnetic compatibility of the RSI shall conform to MIL-STD-461, except during ignition. Equipment class is tactical non-CE, class IIC.

3.5.11 RSI life. The RSI shall have not less than 600 candelas per square millimeter luminance at the end of 300 hours of operation.

3.6 Identification marking. The RSI shall be identified and marked in accordance with MIL-M-13231.

3.7 Workmanship. Workmanship shall be in accordance with requirement 9 of MIL-STD-454. The quartz envelope shall be free of scratches, bubbles, striations, or blemishes.

3.8 Screw threads. Screw threads shall conform to National Bureau of Standards Handbook H-28.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified by the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Parts and materials. The supplier is responsible for insuring that parts and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards as specified herein.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. First article (see 4.3)

b. Quality conformance inspection (see 4.5)

c. Inspection for preparation for delivery (see 4.7)

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4.3 First article. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.3.1 Inspection. Each RSI shall be examined in accordance with Table I. One or more defects shall be cause for rejection of that RSI.

TABLE I Inspection

Defects	Requirement paragraph
MAJOR	
101. Absence of any part or component	3.2
102. Dimensions and tolerances not as specified	3.2
103. Absence or incorrectness of identification marking	3.6
104. Workmanship not as specified	3.7
105. Screw threads not as specified	3.8

4.3.2 Tests. Upon successful completion of the examination specified in 4.3.1 the first article RSI's shall be subjected to the tests in Table II. Tests shall be conducted in the order listed. Failure of any test shall be cause for rejection.

4.3.3 First article failure criteria. Rejection of any RSI during examination or test (4.3.2) shall be considered first article failure.

TABLE II Inspection

Inspection	Requirement paragraph	Test paragraph
Quartz bulb	3.5.1	4.6.1
Ignition and operation	3.5.2	4.6.2
Luminance	3.5.3	4.6.3
Mechanical shock	3.5.4	4.6.4
Vibration	3.5.5	4.6.5
Low temperature storage	3.5.6	4.6.6.1
Low temperature operation	3.5.7	4.6.7.1
High temperature storage	3.5.6	4.6.6.2
High temperature operation	3.5.7	4.6.7.2
Altitude	3.5.8	4.6.8
Altitude transportation	3.5.9	4.6.9
Electromagnetic compatiblity	3.5.10	4.6.10
RSI life	3.5.11	4.6.11

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4.4 Inspections covered by subsidiary documents. The RSI's shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order. See 4.3.1 and Table I.

4.5 Quality conformance inspection.

4.5.1 Examination. Each RSI shall be examined as specified in Table I. Presence of one or more defects shall be cause for rejection of that RSI.

4.5.2 Tests.

4.5.2.1 Group A Inspection. Each RSI shall be tested in accordance with Table III. Failure of any test shall be cause for rejection of that RSI.

TABLE III Group A Inspection

Inspection	Requirement paragraph	Test paragraph
Quartz bulb	3.5.1	4.6.1
Ignition and operation	3.5.2	4.6.2
Luminance	3.5.3	4.6.3

4.5.2.2 Group B Inspection. Group B inspection shall be conducted on RSI(s) selected from units which have passed the examination and tests in 4.5.1 and 4.5.2.1. Sampling of lots shall be in accordance with MIL-STD-105, AQL 6.5 percent defective, inspection level S-2. Tests shall be conducted in accordance with Table IV, and in the order listed.

TABLE IV Group B Inspection

Inspection	Requirement paragraph	Test paragraph
Mechanical shock	3.5.4	4.6.4
Vibration	3.5.5	4.6.5

4.5.2.3 Group C Inspection. This inspection shall consist of the tests specified in Table V and shall be performed on sample units that have been subjected to and met Group A and B inspection. The tests shall be conducted in the sequence listed. Sample units shall be selected in accordance with 4.5.2.3.1.

4.5.2.3.1 Sampling for inspection. Two sample RSI's shall be selected at random for all inspections in Table V from the monthly specified quantity (see 6.2) produced or each 100 RSI's produced, whichever occurs first. The first sample selected shall be at the start of the contract from the first quality conformance inspection production lot.

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TABLE V Group C Inspection

Inspection	Requirement paragraph	Test paragraph
Low temperature storage	3.5.6	4.6.6.1
Low temperature operation	3.5.7	4.6.7.1
High temperature storage	3.5.6	4.6.6.2
High temperature operation	3.5.7	4.6.7.2
Altitude	3.5.8	4.6.8
Altitude transportation	3.5.9	4.6.9

4.5.2.3.2 Group C failures. Actions required relative to Group C failures shall be as specified in the contract or order (see 6.2).

4.5.2.4 Group D Inspection. This inspection shall be performed on RSI's from lots that have passed Group A inspection. The inspection shall be as specified in Table VI. Three RSI's from every 300 produced, or portion thereof, or from each 3 months production, whichever occurs first shall be selected for inspection (see 6.2).

TABLE VI Group D Inspection

Inspection	Requirement paragraph	Test paragraph
RSI life	3.5.11	4.6.11

4.5.2.4.1 Actions required relative to Group D failures shall be as specified in the contract or order (see 6.2).

4.6 Test methods. Unless otherwise specified in the individual tests, the RSI shall be operated in the horizontal position, surrounded by a quartz or Pyrex tube 60 mm, ± 1.0 mm inside diameter by 80 mm, ± 2 mm long. An air flow of 37, ± 5 cubic feet per minute shall be provided at the cathode end of the RSI and shall flow from the cathode end to the anode end. The air temperature shall be plus 40°C, ± 5°C at the cathode end of the RSI when measured with a thermocouple located as shown in Figure 1. The air outlet at the anode shall be restricted to provide a static pressure of 4, ± 0.5 centimeters when measured with a water manometer. The RSI shall be considered as stabilized for all light measurements after 5 minutes operation with cooling airflow. Unless otherwise specified in the individual tests, the RSI shall be operated with the circuit as specified in Figure 2 and a power supply voltage of 28.0 vdc, ± 0.5 volts. Unless otherwise specified all tests shall be performed at room temperature (see 6.3). During all cooling, air shall be applied within 1 second after lamp is ignited.

4.6.1 Quartz bulb. The optical quality of the quartz bulb shall be tested on a shadowgraph or optical transit as follows and the test data recorded. (The RSI shall not be operated during the test.)

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a. A fixture shall be used for mounting the RSI with the electrodes in a horizontal attitude. The fixture shall permit rotation of the RSI about the centerline of the electrodes. The fixture shall also be capable of being rotated in azimuth about a point, formed by the apex of the conical surface of the cathode. The RSI shall be positioned in the fixture so the apex of the conical surface of the cathode is the same vertical centerline as the fixture rotates about in azimuth and so the optical line of sight through the shadowgraph or optical transit is perpendicular and passes through the longitudinal axis of the RSI.

b. Rotate the RSI 360° about its longitudinal centerline without changing the azimuth position.

c. Rotate the RSI 30° in a clockwise direction in azimuth around the vertical centerline through the apex of the conical surface of the cathode and perpendicular to the longitudinal axis and repeat step b above.

d. Rotate the RSI in a counterclockwise direction 22-1/2° around the same centerline as in step c and repeat b above.

e. Repeat step d in 22-1/2° increments for a total of 67-1/2°. Any apparent shift of the cathode more than 0.5 millimeter radius shall constitute failure of this test.

4.6.2 Ignition and operation. The RSI shall be connected as specified in Figure 2 and shall be ignited and operated under the following conditions:

a. The ionization voltage shall be generated by a spark-gap igniter circuit having a frequency of 5 megahertz damped wave output and supplying an open circuit potential of 30 to 40 kilovolts a.c. peak, and a duration of 0.2 microsecond for the first cycle and a total time of 1 microsecond. This potential will be applied to the RSI from the cathode side (series injected), with the anode radio frequency (R.F.) by-passed to ground with a 0.02 microfarad disc ceramic capacitor. The total number of R.F. bursts shall not exceed 150.

b. The boost circuit consisting of 2 direct current supplies, one having an open circuit voltage of 100 vdc and a short circuit current of 2.0 amperes, a second supply having an open circuit voltage of 58 vdc and a short circuit current of 40 amperes, shall be connected as shown in Figure 2. Prior to performing steps c and d below, each RSI shall be operated for a 2 hour period in the circuit with a power supply voltage of 28.0 vdc, $\pm$ 0.5 volts, after which the RSI shall be turned OFF and allowed to stabilize at room ambient temperature for a period of at least one hour.

c. Turn ON the main power supply and adjust the open circuit voltage to the low voltage as specified in 3.5.2. Ignite the RSI and after a minimum of 5 minutes of operation, measure and record the RSI operating voltage and current.

d. Adjust the main power supply to 28 volts dc, plus or minus 0.5 volts. Ignite the RSI and after a minimum of 5 minutes of operation, measure and record the RSI operating voltage and current.

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e. Failure of the RSI to ignite within 3 seconds or failure to comply with 3.5.2 shall constitute failure of this test.

4.6.3 Luminance. The RSI shall be supported in a horizontal position and operated with a power supply voltage of 28.0 vdc, ± 0.5 volts. The average luminance of the arc over a circular area 1.0 millimeter in diameter centered on the optical axis of the RSI shall be not less than 840 candelas per square millimeter using a lens with a stop of known area and a photocell corrected to a photopic response curve. The photocell shall be calibrated with a tungsten lamp of 1000 watts operated at a color temperature of 2870° K, plus or minus 50° K. Measured luminance of less than 840 candelas per square millimeter shall constitute failure of this test.

4.6.4 Mechanical shock. The RSI shall be rigidly mounted to a shock table. Throughout this test the RSI shall be operated with the driving circuit as specified in Figure 2 and a power supply voltage of 28.0 vdc, ± 0.5 volts. The RSI shall be subjected to 12 shock impacts. Measurements of shock level shall be taken at the attach point for the anode.

a. Six shocks shall be applied at 70 g's, ± 5 g's with the optical axis parallel to the direction of shock, with the time duration of each to be 6.0 milliseconds, plus or minus 0.5 millisecond.

b. Repeat step a with the optical axis perpendicular to the direction of shock. Failure of the RSI to comply with 3.5.4 shall constitute failure of this test.

4.6.5 Vibration. The RSI shall be installed in a test fixture mounted on a vibration table and shall be operated at 1000 watts. Precautions shall be taken to prevent magnetic fields from influencing stability of the arc. The RSI shall be operated during the entire vibration period as defined below.

a. Vibrate the RSI along the axis of the lamp electrodes at a constant peak acceleration of 16.0 g's, plus or minus 0.5 g and with frequency varying linearly from 21 Hz to 26 Hz. The frequency shall change at the rate of 1.0 Hz per minute for a total sweep duration of not less than 5 minutes.

b. Vibrate the RSI along the axis of the lamp electrodes at a constant peak acceleration of 8.0 g's, plus or minus 0.5 g and with frequency varying linearly from 130 Hz to 150 Hz. The frequency shall change at the rate of 4.0 Hz per minute for a total sweep duration of not less than 5 minutes.

c. Vibrate the RSI in a direction perpendicular to the axis of the lamp electrodes at a constant peak acceleration of 16.0 g's, plus or minus 0.5 g and with frequency varying linearly from 29 Hz to 34 Hz. The frequency shall change at the rate of 1.0 Hz per minute for a total sweep duration of not less than 5 minutes.

d. Vibrate the RSI in a direction perpendicular to the axis of the lamp electrodes at a constant peak acceleration of 10.0 g's, plus or minus 0.5 g and with frequency varying from 70 Hz to 90 Hz. The frequency shall change at the rate of 4.0 Hz per minute for a total sweep duration of not less than 5 minutes.

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Any evidence of damage or failure of the RSI to maintain operation shall constitute failure of this test.

4.6.6 Storage Temperature.

4.6.6.1 Low temperature storage. The RSI shall be placed in a test chamber and exposed to an ambient temperature of minus 62°C and maintained for a period of 6 hours. At the conclusion of the exposure period, the RSI shall be examined. Any evidence of damage shall constitute failure of this test.

4.6.6.2 High temperature storage. The RSI shall be stored in a test chamber at an ambient temperature of plus 68°C with humidity controlled at 85 to 95 percent for 6 hours. At the conclusion of the storage period the RSI shall be ignited and operated for 5 minutes, turned OFF and allowed to cool. Failure to ignite and maintain ignition or any evidence of damage shall constitute failure of this test.

4.6.7 Operating temperature.

4.6.7.1 Low temperature operation. The RSI, installed in a test fixture shall be placed in a test chamber and cold soaked at minus 54°C for 2 hours. The RSI shall then be ignited and shall be operated with cooling air for 3 minutes. The RSI shall then be turned OFF and allowed to cool for 12 minutes in the test fixture. The above cycle, not including the 2-hour cold soak, shall be repeated 4 times. Any evidence of damage or failure of the RSI to operate shall constitute failure of this test.

4.6.7.2 High temperature operation. The RSI shall be mounted in a test fixture. The RSI and fixture shall be placed in a test chamber and subjected to 2 cycles of exposure with the relative humidity maintained in the chamber at 90 percent, $\pm$ 5 percent at all times. Each cycle shall consist of the following:

a. Two hours at plus 52°C.

b. Decrease temperature within 1 hour to plus 23°C.

c. Increase temperature within 1 hour to plus 52°C.

d. Maintain plus 52°C for 2 hours.

Upon completion of the 2 cycles the RSI shall be ignited and operated at plus 52°C ambient temperature. Failure of the RSI to ignite within 3 seconds and operate within the limits specified in 3.5.2 shall constitute failure of this test.

4.6.8 Altitude. The RSI shall be tested in accordance with Procedure II, Method 500 of MIL-STD-810, except the temperature at 10,000 feet shall be plus 35°C. Ignite and operate the RSI continuously for 1 hour. Failure of the RSI to ignite and maintain ignition within 3 seconds and operate for 1 hour, or evidence of damage shall constitute failure of this test.

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4.6.9 Altitude transportation. The RSI shall be placed in a test chamber and the air pressure in the chamber shall be reduced to the value corresponding to 40,000 feet above sea level and a temperature of minus 62°C. The chamber shall be maintained at this pressure and temperature for 4 hours. At the end of 4 hours, the chamber air pressure shall be returned to the value corresponding to sea level and the temperature to room temperature. The RSI shall be ignited and ignition maintained for 5 minutes, then turned OFF and allowed to cool. Failure to ignite or any evidence of damage shall constitute failure of this test.

4.6.10 Electromagnetic compatibility. The RSI's shall be tested in accordance with MIL-STD-462. Failure of the RSI to meet the requirements of 3.5.10 shall constitute failure of this test.

4.6.11 RSI life. The RSI shall be ignited and operated for 300 hours under the following conditions:

a. Ignite and operate for 50 minutes with a driving circuit as specified in Figure 2 and a power supply voltage of 28.0 vdc, ± 0.5 volts.

b. Turn OFF the RSI for 10 minutes.

c. Repeat step a and b until 90 operating hours have been accumulated and perform tests specified in 4.6.2 steps a through d. The accumulated time on previous tests shall be considered a part of this 90 hours.

d. Repeat steps a and b until 300 operating hours have been accumulated. At the end of the 300 hours the RSI shall be ignited and operated as specified in 4.6.2 steps a through d.

e. Perform luminance test as specified in 4.6.3.

Failure of an RSI to ignite as specified, luminance of less than 600 candelas per square millimeter, voltage not within limits specified in 3.5.2 or damage to the RSI shall constitute failure of this test.

4.7 Inspection of preparation for delivery. Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified. (see 6.2)

5.1.1 Level A.

5.1.1.1 Cleaning. Radiation Source, Infrared for Searchlight, Infrared, AN/VSS-3, shall be cleaned in accordance with process C-1 of MIL-P-116.

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5.1.1.2 Drying. Radiation Source, Infrared for Searchlight, Infrared, AN/VSS-3, shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application. None required.

5.1.1.4 Unit packaging. Each Radiation Source, Infrared for Searchlight, Infrared, AN/VSS-3, shall be individually packaged in accordance with method II of MIL-P-116 as follows: Cushion each radiation source, infrared on all surfaces with pads of polyurethane conforming to MIL-P-26514 designed to protect all projections and absorb the shock of impact in handling and transit. Place the cushioned item within a close-fitting box conforming to MIL-B-43014, Form I, style II. Closure shall be as specified in the appendix of the box specification.

5.1.2 Level C. Radiation Source, Infrared for Searchlight, Infrared, AN/VSS-3, shall be preserved and packaged in a manner that will afford adequate protection against physical and environmental damage during shipment, handling and limited storage in transit.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2). Shipping containers for all levels shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the containers or their contents.

5.2.1 Level A.

5.2.1.1 Consolidation. A quantity of Radiation Source, Infrared for Searchlight, Infrared, AN/VSS-3, packaged as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-636, type CF, class weather-resistant. Box closure shall be as specified in the appendix of the box specification. To facilitate palletization, fiberboard boxes shall be uniform in size and contain equal quantities of the packaged items to the greatest extent practicable.

5.2.1.2 Palletized load. A quantity of containers, packed as specified in 5.2.1.1, shall be placed on a pallet, load type 1, conforming to MIL-STD-147 except that the pallet shall be softwood conforming to NN-P-71, type IV, size 2. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3s. The load shall be "bonded" to the pallet by strapping.

5.2.1.3 Less than palletized load. When quantities per destination are less than a pallet load, the containers packed as specified in 5.2.1.1 shall be waterproofed, with tape conforming to PPP-T-76, in accordance with the taping requirements of the appendix of the box specification. A quantity of the water-proofed containers shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the

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weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specifications, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type 1, class B.

5.2.2 Level B.

5.2.2.1 Consolidation. A quantity of Radiation Source, Infrared for Searchlight, Infrared, AN/VSS-3, packaged as specified in 5.1, shall be packed as specified in 5.2.1.1.

5.2.2.2 Palletized load. A quantity of containers, packed as specified in 5.2.3.1 shall be palletized as specified in 5.2.1.2.

5.2.2.3 Less than palletized load. When quantities per destination are less than a pallet load, the containers packed as specified in 5.2.2.1 shall be reinforced by pressure-sensitive filament tape conforming to PPP-T-97, type IV as specified in the appendix of the box specification. No further packing shall be required.

5.2.3 Level C.

5.2.3.1 Consolidation. A quantity of Radiation Source, Infrared for Searchlight, Infrared, AN/VSS-3, shall be packed as specified in 5.2.1.1, except that the fiberboard boxes should be class domestic.

5.2.3.2 Palletized load. A quantity of containers, packed as specified in 5.2.3.1, shall be palletized as specified in 5.2.1.2, except that the fiberboard cap shall be class domestic.

5.2.3.3 Less than palletized load. When quantities per destination are less than a pallet load, the containers packed as specified in 5.2.3.1 shall be used as the shipping container. No further packing shall be required.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The RSI is intended for use as a light source in a Searchlight, Infrared, AN/VSS-3, 50 - 60 amperes.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

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b. Quantity and schedule for first article testing for all line items (see 4.3).

c. Monthly rate of delivery (see 4.5.2.3.1).

d. Level of preservation and packaging and level of packing required (see 5.1 and 5.2).

e. Actions required relative to Group C and D failures (see 4.5.2.3.2 and 4.5.2.4.1).

6.3 Definitions. For the purpose of this specification the following definitions apply.

6.3.1 Damage. Damage is defined as deformation, loosening, breakage, corrosion, change of fit of any component or part or failure to operate.

6.3.2 "g". g is defined as an acceleration or deceleration of 32.17 feet per second per second.

6.3.3 Room temperature. Room temperature is defined as plus 23°C, plus or minus 3°C.

6.3.4 Lot size. A lot is defined as those RSI's successfully completing Group A test for any one month.

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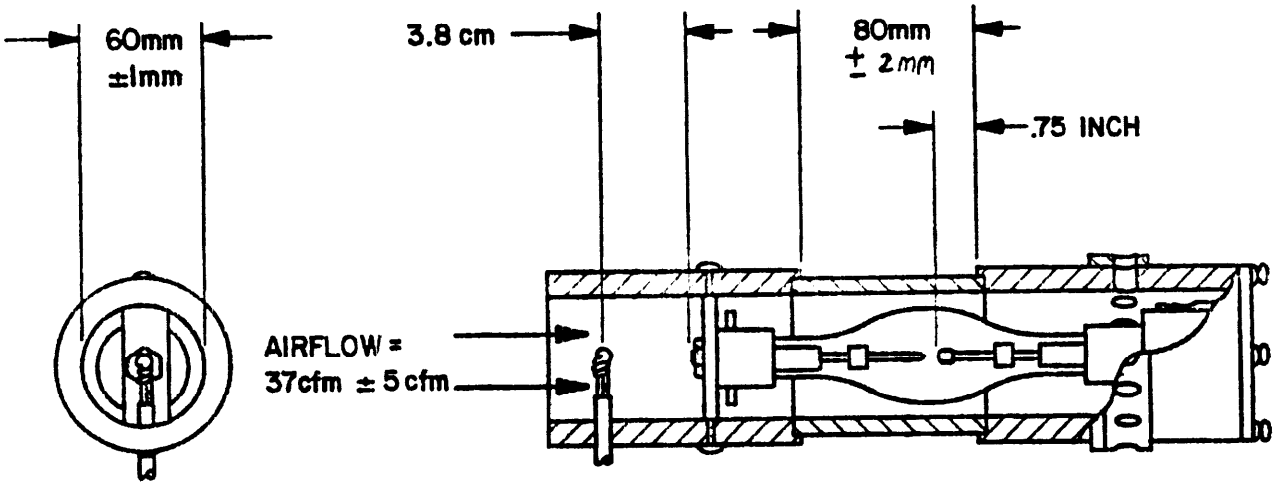
Army-EL
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PREPARING ACTIVITY:

Army-EL

Project No. 5855-0025

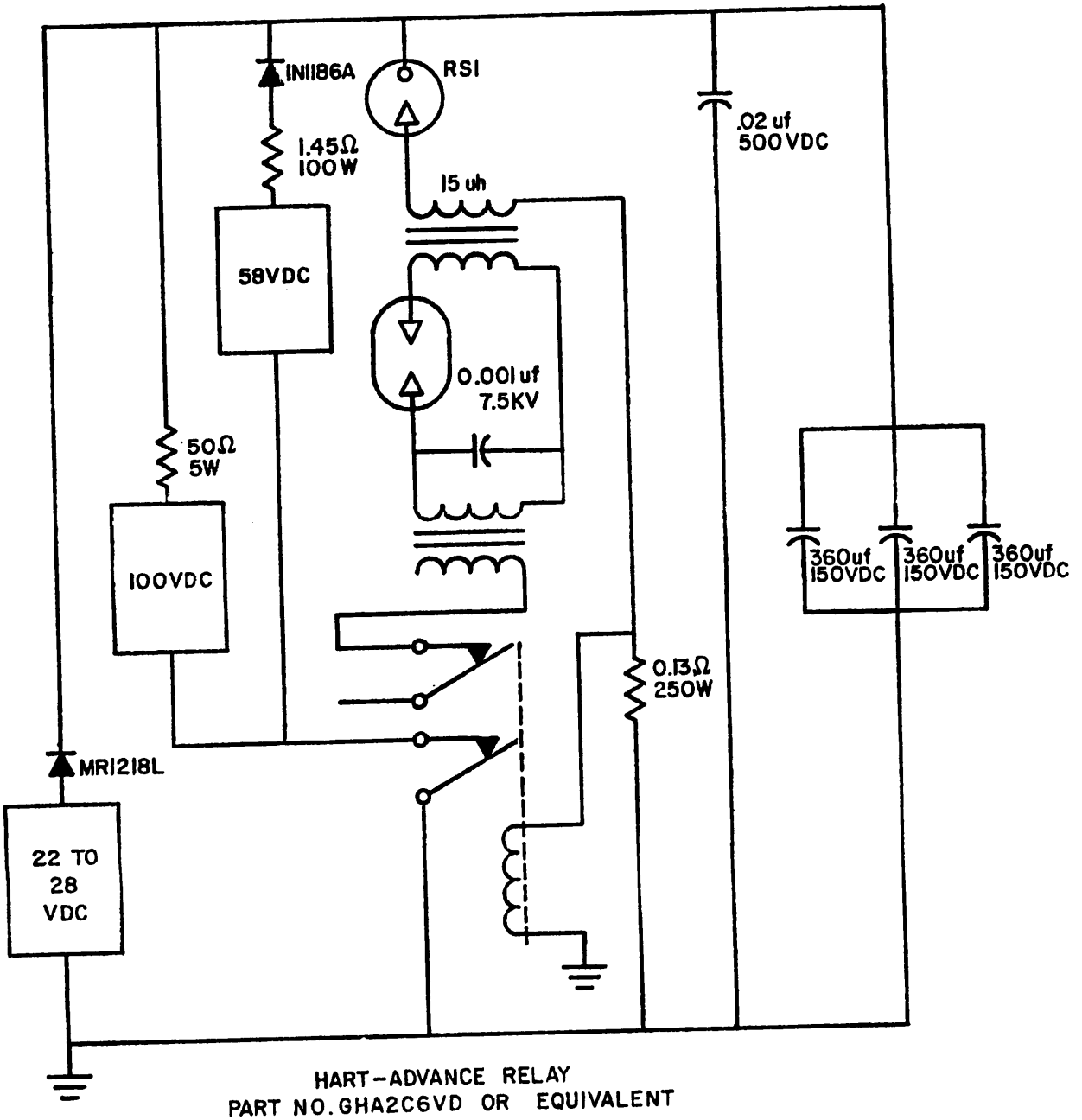
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RSI Airflow

FIGURE 1

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HART-ADVANCE RELAY
PART NO. GHA2C6VD OR EQUIVALENT
Test Circuit for RSI

FIGURE 2

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-R-55344B RADIATION SOURCE, INFRARED FOR SEARCHLIGHT, INFRARED, AN/VSS-3

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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